

Information on the Numerical Analysis module

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Office hours (ricevimento): Friday 9–11 or by appointment.

Lectures Alternating with Prof. Antonio Frangioni.

Irregular schedule, especially at the beginning.

Content

Prof. Frangioni's part: solve all the problems:

$$\min_{x \in S} f(x).$$

This part: solve a very specific problem:

$$\min_{x \in \mathbb{R}^n} \|Ax - y\|^2. \quad (\text{LS})$$

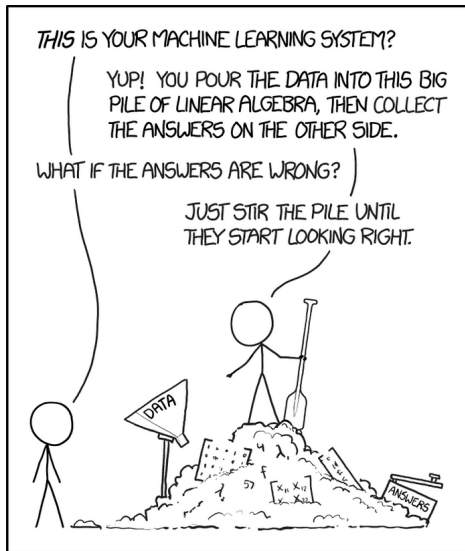
“Solve a linear system, or find the best approximation
of the solution in a very specific norm”

Content (cont.)

Why is this still relevant? It is a problem we can understand well:

- ▶ Specialized algorithms that compute solutions quickly and with very high accuracy, even for very large dimensions (think $\approx 1M \times 1M$ matrices);
- ▶ We are very demanding: can we get the solution up to machine precision 10^{-16} ? If no, why?
- ▶ Often appears as sub-step in more difficult problems;
- ▶ Useful for theoretical understanding: e.g., principal components.

Obligatory XKCD



Books

Books (for this part):

- ▶ Trefethen–Bau, *Numerical Linear Algebra*.
Recommended: we will follow its approach.
- ▶ Demmel, *Applied Numerical Linear Algebra*.
Alternative source for other explanations, exercises, ...
- ▶ Strang, *Linear Algebra and Learning from Data*,
<https://epubs.siam.org/doi/book/10.1137/1.9780692196380>.
- ▶ Eldén, *Matrix Methods in Data Mining and Pattern Recognition*,
<https://epubs.siam.org/doi/book/10.1137/1.9780898718867>.

2. and 3. are accessible for free from within our university network (or via VPN
<https://start.unipi.it/en/help-ict/vpn/>).

Languages

- ▶ Matlab: proprietary language specialized in matrix computations. Clunky language with a big Visual Basic-like IDE, great libraries and syntactic sugar for numerics.
- ▶ Python: great simple language with decent numerics libraries stapled onto it.
- ▶ C/C++, Fortran (yes, that's still a thing): sometimes needed for best performance.
- ▶ Julia: newer, same age as Go / Rust. Stable, but tooling still not perfect. It tries to combine all advantages: syntactic sugar for numerics + libraries + fast loops.

Under the core: the same libraries (BLAS/LAPACK) for basic operations: summing vectors, multiplying matrices. . .

How to install Matlab

Available without charge to Unipi students:

- ▶ Go to <https://unipi.it/matlab>;
- ▶ Log on with your @studenti.unipi.it account;
- ▶ Create a Mathworks account;
- ▶ Download and install.

Suggested: install only the toolboxes that you need. For this course: Matlab, Symbolic Toolbox, Optimization Toolbox.

You may also be interested in: Statistics and Machine Learning Toolbox, Neural Network Toolbox.

Alternative: Cloud Matlab <https://matlab.mathworks.com/>.

Matlab introduction: Command window

```
>> a = 19
```

```
a =
```

```
19
```

```
>> b = 5
```

```
b =
```

```
5
```

```
>> a+b
```

```
ans =
```

```
24
```

```
>> c = 2*a + b; % semicolons suppress output
```

Matlab introduction: Accuracy of operations

Default data type: double (IEEE standard binary64).

Warning: operations are only accurate “up to 16 digits”.

```
>> (1/98) * 98 - 1
```

```
ans =
```

```
-1.1102e-16
```

(exponential notation: stands for -1.1102×10^{-16})

Understanding the effect of these approximations will be an important part of this module.

Numerical linear algebra people are very demanding: we have “simple” problems, and we wish to solve them up to 10^{-16} , or something around there. Not 10^{-8} or 10^{-12} .